ELECTRICAL—ELECTRIC VEHICLE—DIAGNOSTIC TROUBLE CODE (DTC) B2785 AND/OR POWER LIMITED MODE OPERATION—ELECTRIC RANGER AND ELECTRIC POSTAL VEHICLES ONLY	Article No. 01-19-3

FORD: 1998-2001 RANGER

ISSUE

On some Electric Ranger and Electric Postal Vehicles the traction Inverter Module (TIM) cannot identify the type of vehicle it is installed in. A DTC B2785 may be set and the vehicle will enter a Power Limited mode of operation. This may be caused by an incorrect TIM software version.

ACTION

If Code B2785 is present follow the revised diagnostic flow chart. Refer to the following Service Information for details.

SERVICE INFORMATION

The (TIM) contains three software versions. There are separate programs for PbA, NiMH and Postal vehicles. The differences primarily involve regenerative braking schedules, and the max speed for postal vehicles (60mph). When the vehicle is started, the TIM requests that the Battery Control Module (BCM) identify itself as either PbA or NiMH (All postal vehicles are PbA). The TIM then looks for an interconnection between pins C4998-1 and C4998-17 and pins C4999-2 and C4998-28 (Figures 1 and 2). If this connection is seen, the TIM recognizes the vehicle as an Electric Postal vehicle. If the TIM sees one interconnection and not the other, DTC B2785 will be set and the vehicle will enter a Power Limited mode of operation. If the TIM sees neither, the vehicle is assumed to be a Ranger EV.

Condition	Possible Source	Action
DTC B2785 “TIM/Vehicle Wiring Harness Identification”	•Vehicle wiring •TIM module	Go to Pinpoint Test C

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PINPOINT TEST C: DTC B2785 "TIM/Vehicle Wiring Harness Identification"	
Test Step	**Action To Take**
C1 CHECK CIRCUIT X (LB/BK) FOR CONTINUITY •With key OFF, remove connectors C4998 and C4999 and measure the resistance between pins C4999-2 (LB/BK) and C4998-28 •Is the resistance less than 5 ohms?	Yes= GO to C2 No= REPAIR the wires in question
C2 CHECK CIRCUIT X (LB/BK) FOR SHORT TO POWER •With key ON, measure the voltage between pin C4999-2 (LB/BK) and pin C4998-10 (R/Y) •Is the voltage greater than 0 volts?	Yes= REPAIR LB/BK wire short to POWER No= GO to C3
C3 CHECK CIRCUIT X (LB/BK) FOR SHORT TO GROUND •With key OFF, measure the resistance between C4999-2 (LB/BK) and C4998-29 (BK/LB) •Is the resistance less than 5 ohms?	Yes= REPAIR LB/BK wire short to ground No= GO to C4
C4 CHECK CIRCUIT Y (DG/BK) FOR CONTINUITY •With key OFF, measure the resistance between pin C4998-1 (DG/BK) and C4998-17 •Is the resistance less than 5 ohms?	Yes= GO to C5 No= REPAIR the wire in question
C5 CHECK CIRCUIT Y (DG/BK) FOR SHORT TO POWER •With key ON, measure the voltage between pin C4998-1 (DG/BK) and pin C4998-10 (R/Y) •Is the voltage greater than 0 volts?	Yes= REPAIR short to power No= GO to C6
C6 CHECK CIRCUIT Y (DG/BK) FOR SHORT TO GROUND •With key OFF, measure the resistance between C4998-1 (DG/BK) and C4998-29 (BK/LB) •Is the resistance less than 5 ohms?	Yes= Repair DG/BK wire short to ground NO= GO to C7
C7 CHECK FOR INTERMITTENT SHORT TO GROUND/POWER •Reconnect C4998 and C4999 to the TIM module •With key ON, clear the TIM DTCs •Cycle the key from ON then OFF then START •Does DTC B2785 recur after being cleared?	Yes= REPLACE the TIM No= Suspect an intermittent short to POWER or GROUND. Examine circuit wires, pins and connectors for causes of short/open circuit

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 190000, 290000, 390000, 690000

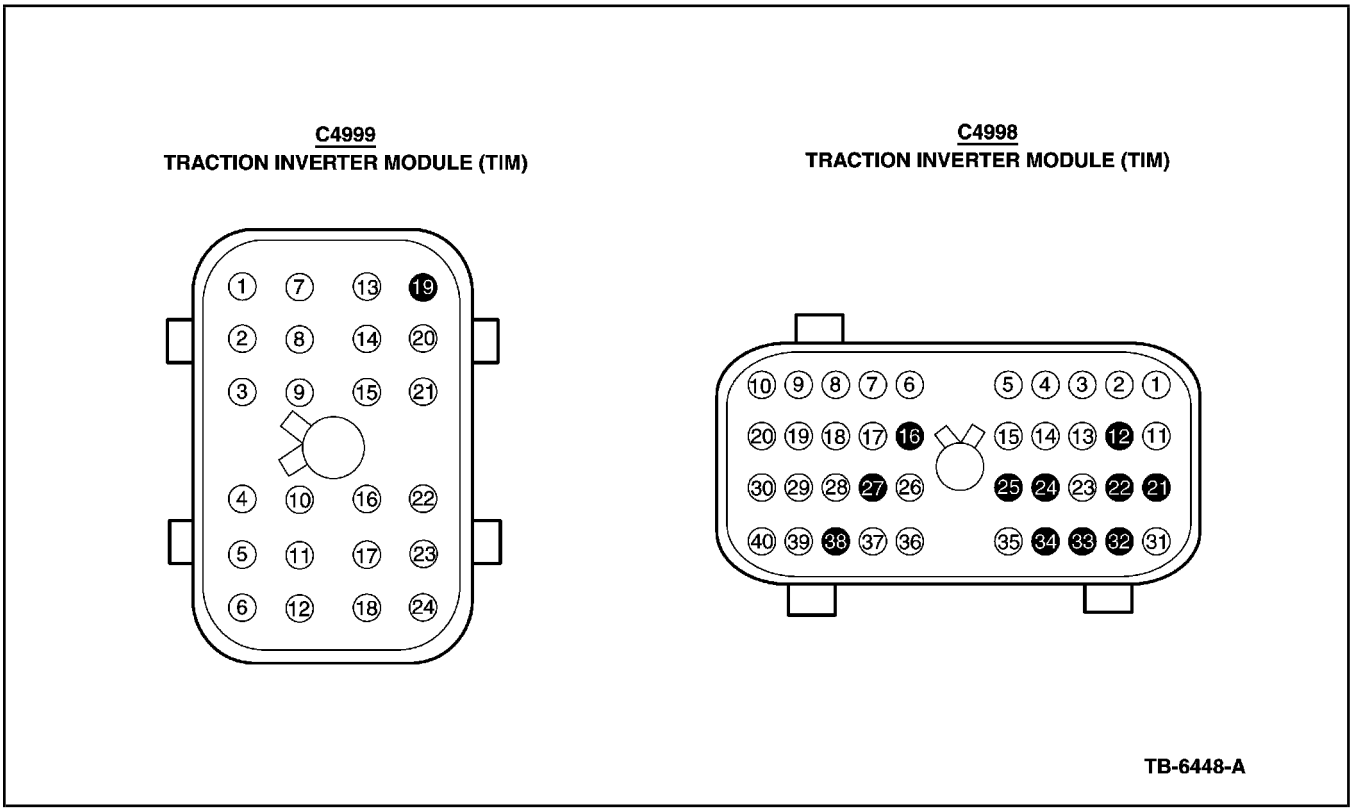

Figure 1 - Article 01-19-3

2001 RANGER ELECTRIC VEHICLE MOTOR/TRANSAXLE CONTROL

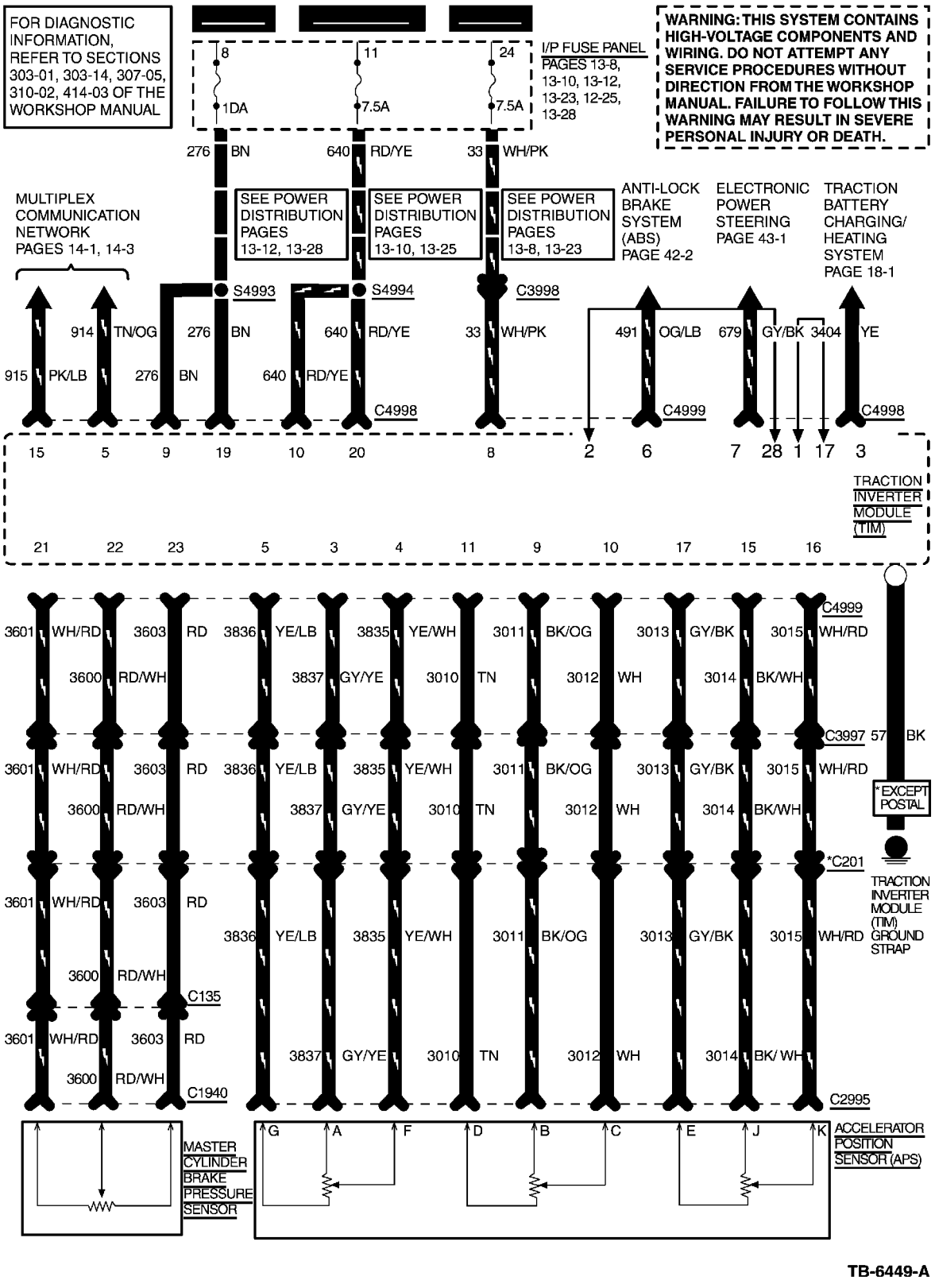

Figure 2 - Article 01-19-3